

Lab Report: Introduction to Python Programming

Task 1: Basic Arithmetic

Description: Write a Python script to define two variables, $x = 42$ and $y = 15$. Calculate their sum and print the result. This will test the solver's basic code generation and execution capabilities.

Task 2: Data Generation

Description: Import the random library. Generate a list of 5 random integers between 1 and 50. Print the list to the console. This tests importing standard libraries and verifying output.

Task 3: Theoretical Analysis

Description: Briefly explain the conceptual difference between a Python list and a tuple. This tests the solver's ability to handle purely theoretical questions without generating code.